

MATTHEW BUCHANAN
CSCI 499 SENIOR PROJECT DEFENSE
DECEMBER 6, 2019

CSCI 499 STATEMENT OF PURPOSE

The purpose of my Senior Project was to create a program that would schedule activities for campers to attend at Camp La Vida. The program needed to allow the staff at Camp La Vida to input the names of the children attending camp and the camp activities the kids wanted to attend, as well as have the ability to create a schedule of the camp activities they wanted to attend for the week. Also, the schedule needed to be made on a daily basis so that certain activities could be toggled on or off depending on the weather. This program needed to be flexible enough so that it could be adjusted as camp progressed and camp activities were adjusted; while this goal was achieved for the purposes of the current iteration, I plan to continue to improve upon this. On top of the main goals that were accomplished upon deployment, there were several enhancements that occurred afterwards to provide a more satisfactory application. These enhancements were an option to input the grades of the children allowing for grade restrictions on certain activities, the option to reassign kids to different activities when needed, the option to choose different types of schedules based on different camp programs, the ability to post the totals attending camp for the week, the ability to post the totals for the camp activities for the day, and the ability to reschedule the camp activities for the day.

CSCI 499 RESEARCH/BACKGROUND

My Senior Project is a people scheduler application made for Camp La Vida. The application was made to replace the scheduler application that they were using as it did not work properly. To code this project, I went through the online Ruby on Rails tutorial at <https://www.railstutorial.org/book> by Michael Hartl. This tutorial walked me through how to build a basic web application using Ruby on Rails. After finishing the tutorial, I was able to build most of the application as it is for Senior Project Defense. I also had to find a tutorial for setting up my development database environment to PostgreSQL; this change was necessary because I used database types that were specific to PostgreSQL. The Rails Tutorial already covered using PostgreSQL in production, so I did not need to research that. I found a very helpful article that showed me how to change the development environment here: <https://www.daveferrara1.com/ruby-in-rails-switch-from-sqlite3-to-postgres/> posted by @daveferrara1.

CSCI 499 HARDWARE/SOFTWARE REQUIRED

This project is a Ruby on Rails web application and as such only requires a computer with internet connection and a web browser (preferably Google Chrome) to run. To develop the application, I used Amazon Web Service's Cloud9; all I needed to run Cloud9 was a laptop with Google Chrome. Using Cloud9 was recommended by the Ruby on Rails Tutorial by Michael Hartl that I used for research. It hosts a Linux development environment that made the Ruby on Rails setup much easier than on a local machine since most things were preconfigured for me.

Log in | Camp La Vida Scheduler x + camp-scheduler-15493.herokuapp.com

CAMP LA VIDA SCHEDULER

Log in

Email

Password (forgot password)

Log in

The Camp La Vida Scheduler Application by Matthew Buchanan

Login Contact Patch Notes Camp La Vida Scheduler App ver. 1.6

Type here to search

This is the login screen. When navigating to the app, this is the first screen that will appear. The rest of the application cannot be accessed unless a user is logged in. After logging in the navbar also has more links to different admin function.

CAMP LA VIDA SCHEDULER

Admin Functions

What would you like to do?

Add a Camper

View the List of Campers

View today's Schedule

Create the next Schedule

Create the next Mother/Daughter Camp Schedule

Totals for Campivities

Irreversible Actions

What would you like to delete/reset?

Delete the list of Campers

Reset today's Schedule

Reset the entire Schedule

This is the admin page and also the home page. This page holds the links to all of the main scheduler functionalities. All of the "Irreversible Actions" are protected from accidental clicks by a confirmation box.

CAMP LA VIDA SCHEDULER

Add camper

Name

Billy Joe 1

Grade level only affects a camper's ability to go to Rock Wall
Anyone that is under 5th grade will not be able to go to Rock Wall

Grade

3

For the following, please enter 1-7 based on the campivities desired by the camper
The number 9 means the camper did not choose that activity and the number 8 will be
ignored by the scheduler

Archery

1

Rifle

2

Canoeing

3

Baking

9

Camping

This is the add camper page. This page is where staff can enter the camper into the database along with their grade and their desired activities. The grade is used to keep young campers from being scheduled for activities that they cannot go to. The bottom of the page has a submit button that posts the data to the database. Each camper preference slot only allows numbers 1-9 as input.

CAMP LA VIDA SCHEDULER

Billy Joe 1's Activities

Grade: 3

- 1 Archery
- 2 Riflery
- 3 Canoeing
- 4 Camping
- 5 Fishing
- 6 Hiking
- 7 Pottery
- 8 Interpretive Movement
- Edit

The Camp La Vida Scheduler Application by Matthew Buchanan

This is the view camper page. This allows staff to view a camper's activities and also provides a link to edit those activities if need be.

Update Camper | Camp La Vida x +camp-scheduler-15493.herokuapp.com/campers/947/edit

CAMP LA VIDA SCHEDULER

HomeHelpAdd CamperView CampersView ScheduleCreate ScheduleAccount

Update camper

Name

Billy Joe 10

Grade level only affects a camper's ability to go to Rock Wall and Camp Out. Anyone that is under 5th grade will not be able to go to Rock Wall or Camp Out.

Grade

10

For the following, please enter 1-7 based on the campivities desired by the camper. The number 9 means the camper did not choose that activity and the number 8 will be ignored by the scheduler.

Archery

9

Rifery

9

Canoeling

9

Baking

9

Canceling

Type here to search

2:29 PM12/6/2019

This is the edit camper page. This allows staff to edit a camper's activities. At the bottom of the page there is a button to submit the edited information to the database.

CAMP LA VIDA SCHEDULER

All Campers

Total number of campers: 10

Totals for Campactivities

Billy Joe 1	Edit	Delete
Billy Joe 10	Edit	Delete
Billy Joe 2	Edit	Delete
Billy Joe 3	Edit	Delete
Billy Joe 4	Edit	Delete
Billy Joe 5	Edit	Delete
Billy Joe 6	Edit	Delete
Billy Joe 7	Edit	Delete
Billy Joe 8	Edit	Delete
Billy Joe 9	Edit	Delete

This is the camper list page. This page displays all of the campers that have been put into the database. This page gives links to the view and edit pages for each camper, as well as an option to delete that camper. For more than 10 campers, this page is paginated with at most 10 campers to each page.

CAMP LA VIDA SCHEDULER

Totals for Campactivities

Total number of campers: 10

Total Campers wanting to go to Archery: 2

Total including optional choices: 3

Total that have already attended: 0

Total Campers wanting to go to Riflery: 2

Total including optional choices: 4

Total that have already attended: 0

Total Campers wanting to go to Canoeing: 2

Total including optional choices: 4

Total that have already attended: 0

Total Campers wanting to go to Baking: 2

Total including optional choices: 3

Total that have already attended: 0

Type here to search

This is the totals page. This page displays the total number of campers in the database and show which activities will be most popular this week.

Create Schedule | Camp La Vida x +camp-scheduler-15493.herokuapp.com/make

CAMP LA VIDA SCHEDULER

Create Schedule

Please select all of the activities that are going to be on for the next schedule

Archery	☐
Rifery	☒
Canoeing	☐
Baking	☒
Camping	☒
Rock Wall	☐
Fishing	☒
Hiking	☐
Pottery	☒

This is the create schedule page. This page allows staff to toggle activities on and off for the creation of each day's schedule. At the bottom of the page, there is a button to submit the activities for the day and create the schedule.

CAMP LA VIDA SCHEDULER

Activities Schedule Slot 1

Toggle Reassign Links On

Unassigned Campers for this Slot
No unassigned campers

Archery

Rifery

Activity Off Today

Canoeling

Activity Off Today

Baking

Activity Off Today

This is the view schedule page. This page lists out the activities and whether or not they are off for the day. If any campers did not get assigned, they will show up at the top of the schedule for manual assignment using the assignment link. Additionally, if necessary, the reassignment links can be toggled on so that any camper can be reassigned.

CAMP LA VIDA SCHEDULER

Games

Activity Off Today

Guitars

Billy Joe 3 | 1

Billy Joe 4 | 1

Billy Joe 5 | 1

Billy Joe 6 | 1

Billy Joe 8 | 1

Billy Joe 9 | 1

Billy Joe 10 | 1

Billy Joe 7 | 1

Billy Joe 2 | 1

Choir

[Continued from above] This is an example of what an activity looks like with campers scheduled for it. The list includes the camper's name as well as the preference they had for this activity. If reassignment links are on, they will appear to the left of the camper's name.

CAMP LA VIDA SCHEDULER

Reassign camper

Billy Joe 1's remaining desired activities

- 1 Archery | 2 Rifle | 3 Canoeing | 4 Camping | 6 Hiking | 7 Pottery | 8 Interpretive Movement

Please select one activity for the camper to be reassigned to

Archery

Current enrollment: 0

Rifle is off today

Canoeing is off today

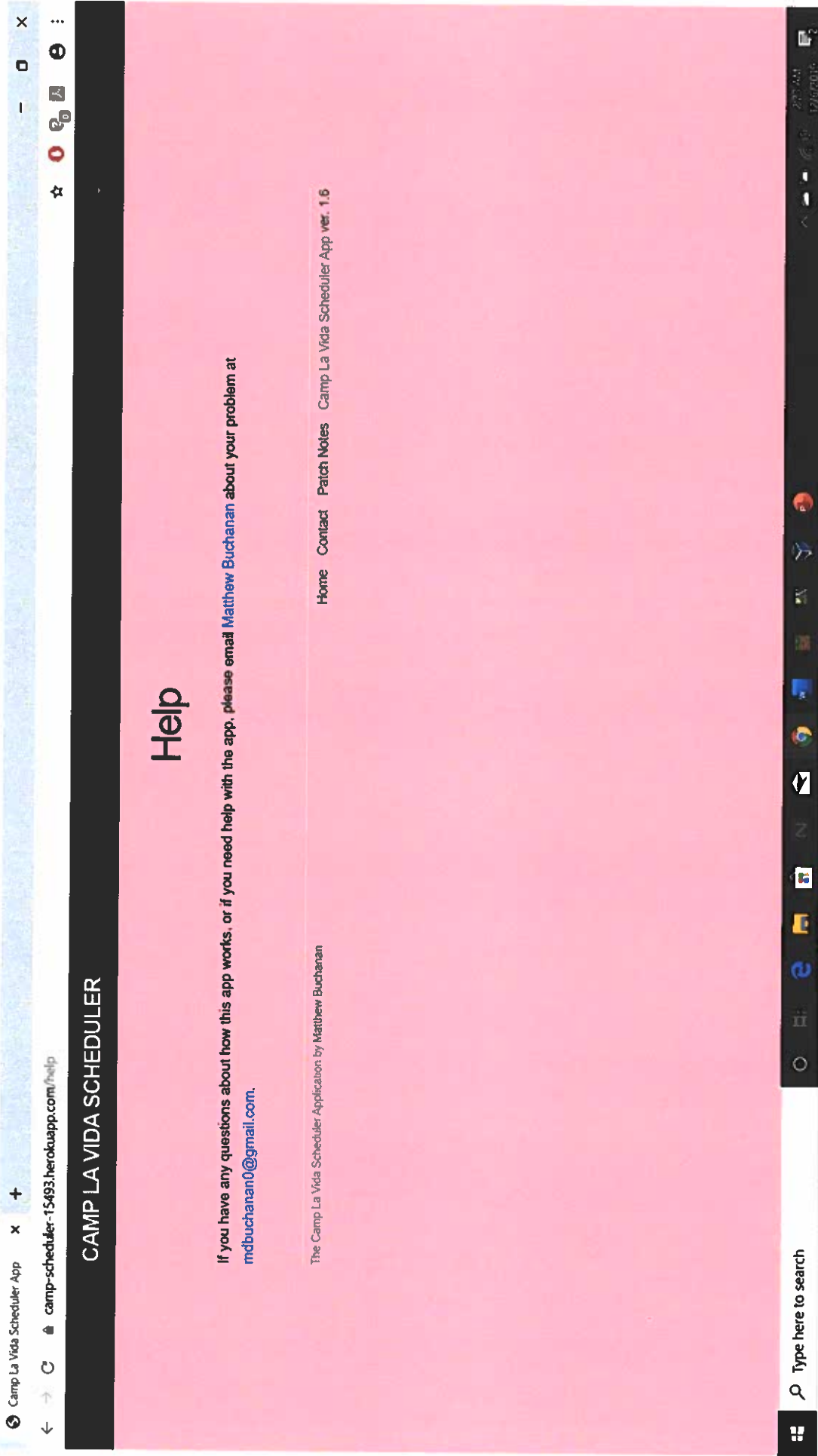
Baking is off today

Camping is off today

Rock Wall is off today

Billy Joe 1 has already had Fishing

This is the reassignment page. This page allows a camper to be reassigned to another activity if the need arises. The page does not allow campers to go to activities that they have already attended or that are turned off for the day. At the bottom of the page there is a button that submits the information, removes the camper from the activity they were in, and adds them to their new activity.



This is the help and contact page. This page gives my contact information so that staff can contact me about the application if needed.

Edit user | Camp La Vida Schedi... x + camp-scheduler-15493.herokuapp.com/users/1/edit

CAMP LA VIDA SCHEDULER

Home | Help | Add Camper | View Campers | View Schedule | Create Schedule | Account

Update your profile

Username

Email

Password

Confirmation

Save changes

The Camp La Vida Scheduler Application by Matthew Buchanan

Home | Contact | Patch Notes | Camp La Vida Scheduler App ver. 1.6

Type here to search

This is the update profile page. This page allows staff to update the information used to log into the program so that a new email and/or password can be entered. This will be useful for staff if the responsibilities of running the program were to be delegated to someone else. The "Username" slot is not used to log in but does show who should be using the program.

CAMP LA VIDA SCHEDULER

Developer Patch Notes and Previews of Future Changes

Version Notes:

7/8/19 Ver 1.6: Added the age limit to also affect the Camp Out Campivity
Added the total number of campers that want to go to each activity along with how many campers have attended each activity
Added a link to the Campivity totals page on the home page and the list of campers
Added an explicit link to the patch notes page in the footer
Changed the name of "Mini Schedule" to "Mother/Daughter Schedule" for better clarity

6/24/19 Ver 1.5: Added a Mother/Daughter camp scheduler that has the same activities but at half capacity to allow mothers to also attend
Added a form that posts to the mini camp scheduler with a new button on the admin page to get to the mini scheduler
Added a total enrollment count to the assign and reassign forms so that the totals can be seen for mini camp scheduling purposes

6/17/19 Ver 1.4: Changed the limit of baking to 15 (was 10)
Added a grade attribute to the camper model's form so that no one under 5th grade may go to Rock Wall
Changed the scheduler, assign form, and reassign form to no longer allow anyone under 5th grade to be in Rock Wall

6/15/19 Ver 1.3: Added a button that toggles the link for any camper to be reassigned

6/11/19 Ver 1.2: Added a button that resets the most recently made schedule
Added a link to the add campers page to the navbar

6/10/19 Ver 1.1: Quality Enhancements:
Added a Reassign link to reassign a camper if the activity doesn't have enough campers
Added tracking for Assign and Reassign pages to no longer allow selecting an activity that a camper has previously had
Added better descriptions and instructions for each form

Type here to search

This is the patch notes page. I included this page to give staff an overview of the changes that I made throughout the summer as they were using the app. Additionally, at the bottom of the page, I have what I plan on doing to improve the app based on staff suggestions/needs and based on what I think would be good enhancements.

TEST PLAN

Test Plan Identifier:

- TEST_PLAN_001

Introduction:

This test plan will ensure that the Camp La Vida Scheduler App will be able to function correctly. To ensure that the app functions correctly, each part will be tested as well as an integration test of the entire app. The log in functionality, the camper list functionality, the scheduler functionality, and the viewability of the schedule, will each be tested individually. Finally, all four will be tested in conjunction. The log in will be tested to ensure that a user will remain logged in until the browser closes or the user logs out. The list of campers will be tested to ensure that campers can be added or deleted and that the list can be cleared all at once. Tests will be run on the scheduler functionality to ensure that it produces an accurate schedule and that schedule items can be added and removed as needed. The schedule will then be tested to ensure that it can be viewed correctly and is presented in an easy to understand way.

References:

- My Senior Project Schedule (listed in full in the Schedule section)
- My Senior Project Design Paper

Features to be Tested:

- User Log In
- User Log Out
- Adding campers to the list of campers
- Removing campers from the list
- Deleting the campers list
- Compiling the Schedule
- Adding items to the Schedule
- Removing items from the Schedule
- Viewing the Schedule

Features Not to Be Tested:

- The styling of the site
 - Functionality comes before aesthetics, if there is time this may be tested
 - Consistency, however, will be tested, such as positional consistency for links

Approach:

- All the tests will be partially automated and partially manual.
- They will use RoR's built in testing feature and also manually testing the app.
 - When able, the built-in test suite will be used to ensure that updating and/or changing the code still passes the test cases

- Manual tests will be done as well closer to the end of the development cycle

Item Pass/Fail Criteria:

- Log In Pass: User can log in and remains so across all pages until logged out
 - Fail: User cannot log in; User does not stay logged in
- Log Out Pass: User can log out by clicking the “log out” link or by closing the browser
 - Fail: User stays logged in after either scenario
- Camper List Pass: Campers can be added, campers can be deleted, and the list can be deleted
 - Fail: Campers cannot be added and/or deleted, the list is not properly deleted
- Schedule Functionality Pass: The schedule is compiled correctly, with no errors
 - Fail: Any errors in the schedule
- Schedule Viewability Pass: The view of the schedule is easy to understand and is printable
 - Fail: The view is cluttered or difficult to read, it is not printable

Suspension Criteria and Resumption Requirements:

- Testing will halt upon deployment to focus on handling any errors that may be found in the field
 - Testing resumes one to two weeks after deployment as new features and/or enhancements are added

Test Deliverables:

- The Test Plan
- Test Cases
 - One for each page
 - One for each form
 - One for each task
- Integration Tests
 - Signup confirmation
 - Login confirmation
 - Logout confirmation
 - Camper addition
 - Camper deletion
 - List deletion
 - Schedule compilation

Test Environment:

- The app will be tested using several different web browsers since it is a web-based app
 - Edge, Chrome, Mobile browsers, possibly Internet Explorer for backwards compatibility

Schedule:

- Beginning Coding: January 2019
- Create a requirements document listing all of the requirements of the system: January 2019
- User Interface Complete: Early April 2019
- Create a complete Test Plan: April 2019

- Camper List Functionality Complete: Late April 2019
- Scheduler Functionality Complete: Late April, Early May 2019
- Fully Coded Beta Version Implementation: Mid May 2019
- Start of Bug Testing: Early May 2019
- Bug Fixes: Late May 2019
- Fully Coded and Tested Implementation: June 1st, 2019
- Field Testing: Summer 2019
- Complete Feedback from Field Testing: August 2019
- Finalize Documentation: October 2019
- Practice run-through of presentation: November 8th, 2019
- Project Defense: November 15th, 2019

Staffing and Training Needs:

- Minimal training may be needed to familiarize staff with the program and how it works
 - An overview of how to navigate the app should suffice

Responsibilities:

- Camp Director and Assistant Director: Add/remove campers to the list and compile the schedule
- Camp Staff: View the schedule and print it out

Risks:

- Unauthorized Login/Use of scheduler
 - Risk Reduction: Require a login with secure passwords and force https
- Forgetting password
 - Risk Reduction: An option to email a new password to the camp email address
- Unauthorized viewing of the schedule
 - Risk Reduction: Login is required to view the schedule

Assumptions and Dependencies:

- Currently (as of 4/20/19) it is assumed that a deployment from Heroku will be sufficient for production purposes
- This app would be dependent on Heroku's servers and the extent of Heroku's free services if that is the case

Approvals:

- Dr. Hayes: Senior Project Construction Advisor

CSCI 499 TEST RESULTS

As described in my test plan, I wrote out several test cases using the built-in testing functionality provided by Ruby on Rails. These tests, detailed in the Test Plan, all passed without issue. Additionally, I also performed manual tests of some of the scheduler functionality since they would be impractical to write out using Ruby on Rails. The reassign and assign links were tested by hand to ensure that each worked properly. Each reassign link had to change the student's current activity to another one and also remove the current one from the camper's list of activities that they have been to. Each assign link had to simply add a camper to an activity. This testing revealed some initial typos but after fixing those, both of these links worked for all cases and had no issues.

CSCI 499 CHALLENGES OVERCOME

There were many challenges that I faced in coding my Senior Project, but I was thankfully able to overcome all of them. First, learning Ruby on Rails completely from scratch was a daunting task; thankfully, Dr. Hayes recommended I use the Rails Tutorial by Michael Hartl which helped tremendously in learning a completely new coding language for me. It took a while for me to get used to working with Ruby on Rails, but now I am very comfortable with it.

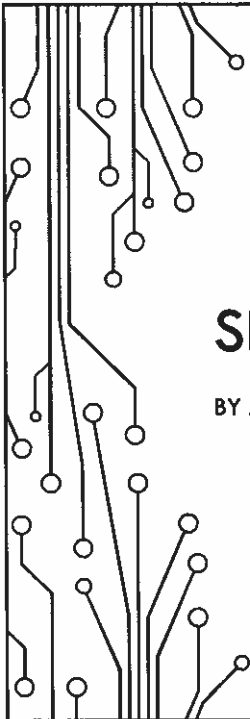
Once I got my basic development application up and running, I ran into a rather large issue that persisted for some time. Basically, I tried to make the daily schedule print out from a Ruby variable where it was stored in one of my page controllers. This method continually worked fine in my development environment, but upon a test deployment to Heroku, my method completely failed me. After much research into what was causing the issue, I found out that the variable I was trying to store the schedule in was a local variable in that page's controller. What that meant for my code was that every time a new GET request was sent to Heroku, the contents of the variable were lost. I also found out the reason that the code worked in my development environment was because the default Rails server and the server used by Heroku were different. The default server for Rails tried to have some semblance of state so that when a page was requested for the second or more time, the variables for the page did not have to be recreated. However, the server that Heroku uses was not state-safe so every new request for a page wiped the values from my variables. After that debacle, I did what I should have probably done from the start but did not think to do: use database entries to hold the schedule. Once I switched over to that, the scheduler functionality ran like it was supposed to at the time.

To implement the database, I wanted to use a feature specific to PostgreSQL, arrays, which sounded easy enough to switch to since my production environment was already configured to use it. However, changing to PostgreSQL in my development environment proved to be a bit challenging. After following a tutorial by @daveferrara1, I thought I had switched fully over to Postgre, but Rails was unable to find my database. Eventually, I found out that I needed to manually start the Postgre service myself since the Cloud9 environment did not have it defaulted to on. Once I started it manually, I was able to use PostgreSQL for my database.

The main challenge that I overcame was a miscommunication with the Assistant Camp Director about how the scheduler needed to function. About a week before camp was set to begin, I met up with the Assistant Director to show her my progress in the application as it was finished from my point of view. When we met up, she was obviously disappointed at the state of the application and told me what she had been expecting from the application. What I had made was a scheduler that was able to take campers from the database and then assign them to activities for the entire week. What she was expecting was for the app to assign campers to activities each day at a time with the option to toggle activities on or off as weather permitted. Her expectations involved basically scrapping what I had for the scheduling functionality and starting over, but I diligently reworked the application that week and was able to get the changes working before camp started. In the future, I will be sure to ask more thorough and detailed questions about what is expected from my work and will check up more frequently to ensure that I am not going in the wrong direction with my work.

CSCI 499 FUTURE ENHANCEMENTS

In the future, I plan to keep adding features and making improvements to the application so that Camp La Vida can continue to use it for camp. First, I plan to make a major scheduler overhaul that should give better priority to camper preferences and hopefully reduce the number of unassigned campers each day. This update would give higher scheduling priority to campers that have not attended their most desired activity. For instance, if there was only one camper on the final day of camp that had still not gone to their most desired activity, then as long as that activity is open, that camper will be guaranteed to be scheduled for that activity. With that update, I also plan to do some backend cleanup that will reduce the overall size of the app which should make load times theoretically faster. Additionally, after the previous enhancements are finished, I plan to change the way camp activities are handled in the back end. This will allow the program to have flexible maximum and minimum limits for activities as well as the ability to add new activities or delete outdated activities. I plan to have most, if not all, of these enhancements finished before camp opens for the summer 2020 season.



SENIOR PROJECT DEFENSE

BY MATTHEW BUCHANAN



STATEMENT OF PURPOSE

- My program was needed to allow the staff at Camp La Vida to schedule their campers' daily activities.
 - The program they had was no good and they needed a replacement.
 - They needed something to create a schedule of the camp activities the campers wanted to attend for the week.
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RESEARCH/BACKGROUND

- Michael Hartl's Ruby on Rails Tutorial
- How to install PostgreSQL tutorial by @daveferrara1

HARDWARE/SOFTWARE REQUIRED

- Web Application
 - Good Internet connection and a good browser (preferably Chrome)
- Development
 - Good Internet connection and a good browser (preferably Chrome)
 - Amazon Web Service's Cloud9

TEST PLAN/RESULTS

- Used Ruby on Rails' built in testing
- Manually tested some scheduler functions
- All tests passed

CHALLENGES OVERCOME

- Learning a new language on my own
- Ruby variables and stateless HTML
- Switching to PostgreSQL
- Miscommunication with the Assistant Director

LIVE DEMONSTRATION

- I apologize in advance for the aggressive amount of pink on the website, it was specifically requested instead of a "boring white" background color.
- URL: <https://camp-scheduler-15493.herokuapp.com/>

FUTURE ENHANCEMENTS

- Scheduler overhaul to prioritize camper preference better
- Code cleanup
- Camp activities change to allow more flexibility
- Planned implementation: Summer 2020